

Cracking the WPL Code - Win Shares for Cricket

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The WPL was established in 2023, with some of the biggest names in the women's game, with the intent of establishing some new ones.

As a new league, there is untapped potential in statistical analysis of players.

Why analyze the WPL?



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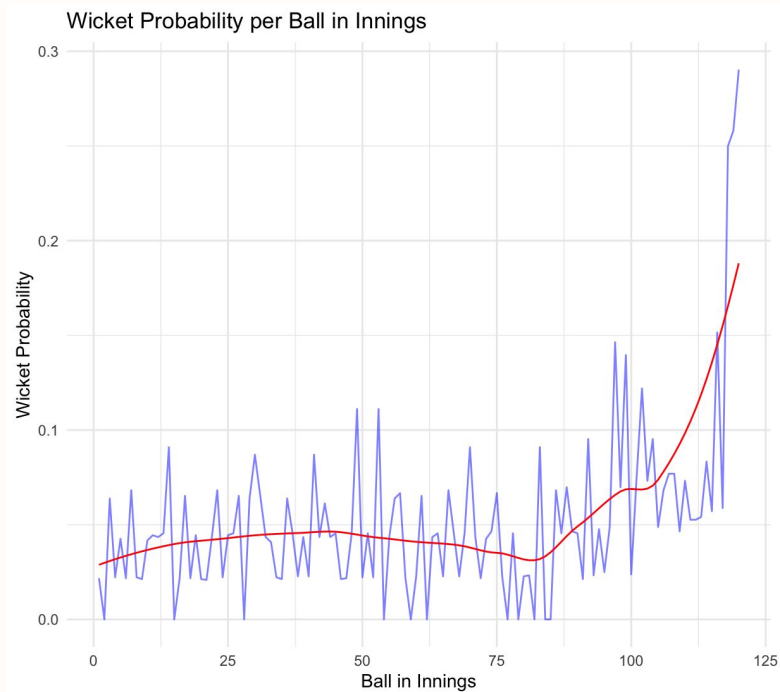
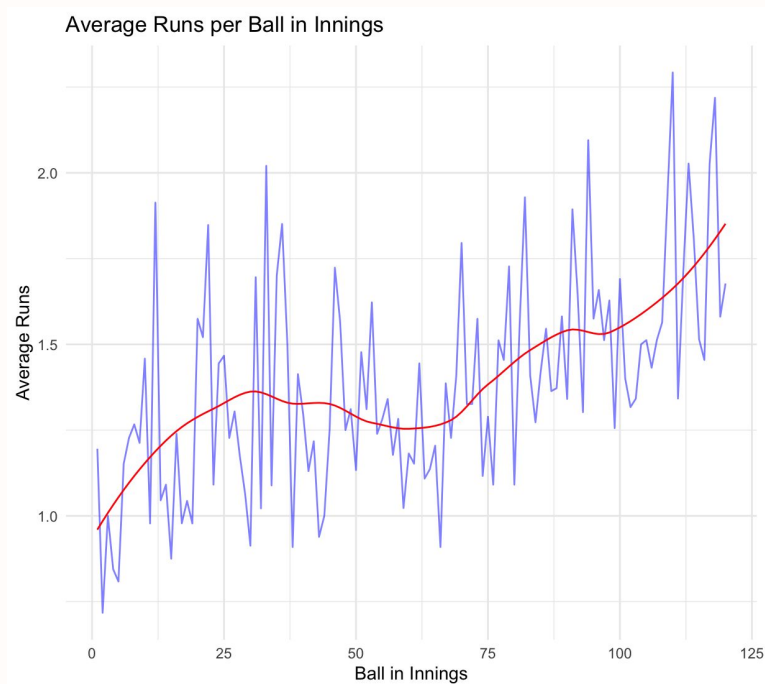
Creating the optimal
team

Working with Ball-by-ball data

Source: Cricsheet, for the WPL 2026 season

Runs on illegal deliveries were counted towards the next legal delivery.

Using Loess for a reasonable average by-ball baseline for both these numbers.



Towards ROE

$$E(R_{ball}) = \text{LOESS}(Runs_{total} \sim Ball_{innings})$$

$$P(W_{ball}) = \text{LOESS}(Wicket_{prob} \sim Ball_{innings})$$

Scale Wickets:

$$W_{base} = \frac{\sum E(R_{ball})}{11 \times \sum P(W_{ball})}$$

Phase adjust:

$$W = W_{base} \times (1 - P(W_{ball}))$$

Defining ROE/WS for Batswomen/Bowlers

$$ROE_{bat} = Runs_{bat} - E(R_{eff}) - (Wicket_{bat} \times W)$$

$$ROE_{bowl} = E(R_{eff}) - Runs_{total} + (Wicket_{bowl} \times W)$$

$$WS_{player} = \frac{z_{bat}}{\sum_{match} |z_{bat}|} + \frac{z_{bowl}}{\sum_{match} |z_{bowl}|}$$

Note: For illegal deliveries (wides, no-balls), the bowler is penalized, but the batswoman is not rewarded.

Team Win Shares

- RCB were the top team in the league, and won the trophy this year in a final against DC. UPW was bottom of the league.
- Especially in DC's case, the win-share metric captures them as a team.

Auction data sourced from ESPNcricinfo.

Team Performance Summary				
	Team	Total WS	Total Spent (₹Cr)	Avg WS/₹
	RCB	4.18	₹12.35	0.13
	DC	1.80	₹12.30	0.40
	MI	-0.11	₹16.60	-0.10
	GG	-2.10	₹14.05	-0.21
	UPW	-4.04	₹14.65	-0.62

Players with Most Win Shares

Interpretation:

Lauren Bell was responsible for **1.89 RCB wins** by herself in the WPL season.

- Note: The top 3 are all on RCB, with a lot of the best value players on DC

Top Performers by Win Shares									
Player	Role	Price	WS	WS/₹	Player	Role	Price	WS	WS/₹
 LK Bell	BOWL	₹0.90	1.89	2.10	 AC Kerr	AR	₹3.00	0.61	0.20
 GM Harris	AR	₹0.75	1.54	2.06	 P Litchfield	BAT	₹1.20	0.60	0.50
 S Mandhana	BAT	₹3.50	1.18	0.34	 H Kaur	BAT	₹2.50	0.45	0.18
 S Ismail	BOWL	₹0.60	1.05	1.76	 NR Sciver-Brunt	AR	₹3.50	0.40	0.11
 NSS Sharma	BOWL	₹0.20	0.97	4.83	 N Shree Charani	AR	₹1.30	0.39	0.30
 SFM Devine	AR	₹2.00	0.89	0.44	 M Mani	AR	₹0.40	0.37	0.93
 S Pandey	AR	₹2.40	0.78	0.33	 CA Henry	AR	₹1.30	0.35	0.27
 L Lee	BAT	₹0.30	0.77	2.57	 RS Gayakwad	BOWL	₹0.40	0.32	0.81
 M Kapp	AR	₹2.20	0.70	0.32	 Vaishnavi Sharma	BOWL	₹0.30	0.28	0.95
 N de Klerk	AR	₹0.65	0.66	1.01	 SR Patil	AR	₹0.60	0.26	0.44

Most Expensive Players

- All-rounders were the most expensive (Nat Sciver-Brunt, Deepti Sharma)
- Balanced with most expensive batswomen (Smriti Mandhana, Harmanpreet Kaur)
- Not a lot of bowlers (most expensive was Asha Shobana)

Player	Role	Price	WS	WS/₹	Player	Role	Price	WS	WS/₹
 S Mandhana	BAT	₹3.50	1.18	0.34	 M Kapp	AR	₹2.20	0.70	0.32
 NR Sciver-Brunt	AR	₹3.50	0.40	0.11	 JI Rodrigues	BAT	₹2.20	0.13	0.06
 A Gardner	AR	₹3.50	-0.67	-0.19	 Shafali Verma	BAT	₹2.20	-0.17	-0.08
 DB Sharma	AR	₹3.20	-0.52	-0.16	 SFM Devine	AR	₹2.00	0.89	0.44
 AC Kerr	AR	₹3.00	0.61	0.20	 MM Lanning	BAT	₹1.90	-0.16	-0.08
 RM Ghosh	BAT	₹2.75	0.13	0.05	 HK Matthews	AR	₹1.75	-0.41	-0.24
 H Kaur	BAT	₹2.50	0.45	0.18	 N Shree Charani	AR	₹1.30	0.39	0.30
 BL Mooney	BAT	₹2.50	-0.54	-0.22	 CA Henry	AR	₹1.30	0.35	0.27
 AB Kaur	AR	₹2.50	-0.83	-0.33	 P Litchfield	BAT	₹1.20	0.60	0.50
 S Pandey	AR	₹2.40	0.78	0.33	 L Wolvaardt	BAT	₹1.10	-0.25	-0.23

Worst Players

- Lots of bigger names who failed to produce in their average match
 - Ashleigh Gardner, Beth Mooney, many members of the WC-winning Indian women's team

Worst Performing Players									
Player	Role	Price	WS	WS/₹	Player	Role	Price	WS	WS/₹
 DJS Dottin	AR	₹0.80	-1.08	-1.35	 BL Mooney	BAT	₹2.50	-0.54	-0.22
 S Rana	AR	₹0.50	-0.92	-1.84	 DB Sharma	AR	₹3.20	-0.52	-0.16
 G Kamalini	BAT	₹0.50	-0.88	-1.75	 Renuka Singh	BOWL	₹0.60	-0.51	-0.84
 AB Kaur	AR	₹2.50	-0.83	-0.33	 CL Tryon	AR	₹0.30	-0.46	-1.53
 A Reddy	AR	₹0.75	-0.81	-1.08	 HK Matthews	AR	₹1.75	-0.41	-0.24
 S Sehrawat	BAT	₹0.50	-0.75	-1.49	 S Sajana	AR	₹0.75	-0.39	-0.52
 H Deol	AR	₹0.50	-0.74	-1.47	 G Voll	BAT	₹0.60	-0.38	-0.64
 A Gardner	AR	₹3.50	-0.67	-0.19	 KP Navgire	BAT	₹0.60	-0.38	-0.63
 KS Gautam	AR	₹0.65	-0.65	-0.99	 NJ Carey	AR	₹0.30	-0.38	-1.25
 S Asha	BOWL	₹1.10	-0.59	-0.54	 NS Prasad	AR	₹0.50	-0.37	-0.73

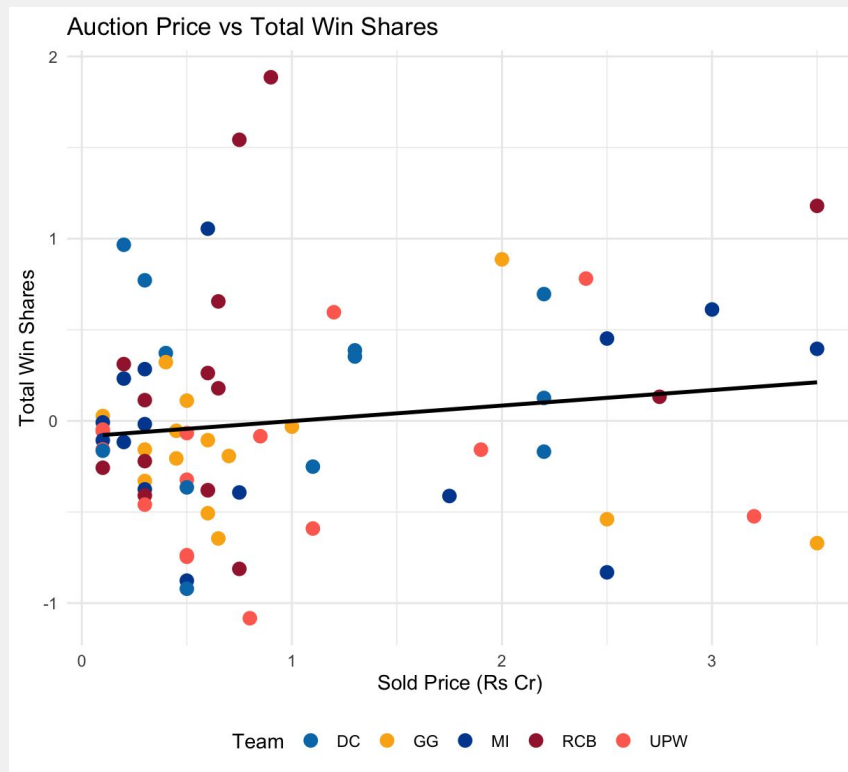
Best Value Players

- Nandani Sharma emerged as an elite pace bowler for her auction price
- Lauren Bell and Lizelle Lee's value show through
- Best value is distributed among player classes rather evenly
- Smriti Mandhana shows she is well worth the hefty price tag

Best Value Players									
Player	Role	Price	WS	WS/₹	Player	Role	Price	WS	WS/₹
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Team Spending Patterns

- You can spot some of the notable data points from before!
- Non-significant F-test p value when regressing ₹Cr (p = 0.21) spend to win-shares, adjusting for role (BAT, BOWL, AR)
 - $R^2 = 0.06$
- **Conclusion: WPL Auction is inefficient**



With WPL Constraints, let's build the best possible team with linear programming!

Our objective is to maximize the sum of win shares with three constraints:

- Maximum 6 overseas players
- 18 player selections
- ₹15 Cr to spend

Mathematically, this is:

$$\max \sum_{i=1}^n WS_i x_i \quad \text{subject to} \quad \begin{cases} \sum_{i=1}^n Price_i x_i \leq 15 & (\text{₹15 Cr Budget}) \\ \sum_{i=1}^n x_i = 18 & (\text{Team Size}) \\ \sum_{i=1}^n Overseas_i x_i \leq 6 & (\text{Max 6 Non-Indians}) \end{cases}$$

The best possible team!

- High value bowlers
 - Nandani Sharma and Lauren Bell again!
- Premium batting is worth it!
 - Phoebe Litchfield, Smriti Mandhana
- General strategy: High value foreigners, expensive foreigners, then domestic diamonds in the rough
 - Nadine De Klerk, Grace Harris
 - Column 2 is all Indian

Optimal Top XI + Reserves									
Budget: 15 Cr Squad: 18 Overseas: Max 6 Min 2 Matches Played									
Player	Role	Price	Total WS	WS/₹	Player	Role	Price	Total WS	WS/₹
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Future
Work +
Thank
you for
listening!

01

Correlation of
win shares
across seasons,
attaching
analysis to
men's IPL

Testing WPL 2023, 2024, 2025 to
see how win shares change for
players within the dataset, also
checking for price

Players to watch: Smriti
Mandhana, Ellyse Perry, Jemimah
Rodrigues

02

Alternative
Definitions of
Win Shares/ROE

Changing to a zero sum
approach, or a more Bayesian
definition (CricWAR by Hassan
Rafique:
[https://github.com/Dazzalytics/
cricWAR_SSAC23](https://github.com/Dazzalytics/cricWAR_SSAC23))

Sort of like comparing the
[basketballreference.com](https://www.basketballreference.com) to Bill
James win shares, eg looking at
Win Shares/Over (like WS/48)

03

Thank You for
Listening!

Hope this inspired you to look at
cricket a little more if you have
not already :)

If you have looked at cricket, I
hope that you have newfound
interest in analytics as well as
more importantly, the emerging
complexities in the women's
game!

Code:
[https://github.com/ArchithShar
ma/WPL_Winshares](https://github.com/ArchithSharma/WPL_Winshares)